

Joel Mugenyi

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joelmugenyi

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Professional Summary

Energy systems researcher and PhD candidate specializing in electricity demand, affordability, and household consumption modeling, with expertise in econometrics and data science applied to large-scale datasets. Experienced in collaborating with electricity regulators, government agencies, and non-government partners to address grid reliability, integrate clean energy into utility systems, and advance affordable off-grid electricity solutions for low-income communities.

Experience

- 2022 – present **Research Assistant:** [Quadracci Sust. Energy Lab](#), Columbia University (NYC, USA).
 - Research on electricity demand, grid reliability, and energy access
- 2024 **Energy Modelling Consultant:** [Equatorial Power](#), Kampala, Uganda.
 - Designed household socio-economic and productivity survey instruments.
- Energy Fellow:** [Center on Global Energy Policy](#), Columbia University (NYC, USA).
 - Collaborating with senior research scholars and participating in policy seminars.
- 2023 **Grid Reliability Consultant:** [Electricity Regulatory Authority](#), Kampala, Uganda.
 - Developed an NLP model to analyze outage reports.
- 2019 – 2022 **Research Engineer:** [eGUIDE Initiative](#), UMass-Amherst (Amherst, USA).
 - Designed and deployed data-driven electricity demand and irrigation projects.
- 2018 – 2019 **R&D Associate:** [PowerGen Renewable Energy](#), Nairobi, Kenya.
 - Developed smart metering and techno-economic strategies for mini-grids.
- 2017 **Maintenance Engineer:** [Eskom Generation Limited](#), Jinja, Uganda.
 - Monitored the performance and operation of ten 18 MW hydroelectric turbines.

Education

- 2022 – 2026* **Ph.D., Earth & Environmental Engineering** Columbia University
Thesis: Affordable Energy Systems in Low-Income Areas
- 2017 – 2019 **M.Sc., Electrical & Computer Engineering** Carnegie Mellon University
Concentrations: Data Science; Off-grid Energy
- 2012 – 2016 **B.Sc., Electrical Engineering** Makerere University
Specializations: Power Systems; Machines; Protection

Teaching

- Fall 2024 **Teaching Assistant:** Solid & Hazardous Waste Management, Columbia University.
- Fall 2018 **Teaching Assistant:** Photovoltaic Systems Engineering, Carnegie Mellon University.
- Spring 2018 **Teaching Assistant:** Software Defined Radios, Carnegie Mellon University.

* Expected graduation.

Research Publications

Journal Articles

- 1 **J. Mugenyi** and V. Modi, "Survey bias and the poor: How survey responses overstate electricity spending," *Energy for Sustainable Development*, vol. 90, p. 101 876, ISSN: 0973-0826.  DOI: [10.1016/j.esd.2025.101876](https://doi.org/10.1016/j.esd.2025.101876).
- 2 L. Masselus, J. Ankel-Peters, G. Gonzalez Sutil, V. Modi, **J. Mugenyi**, A. Munyehirwe, N. Williams, and M. Sievert, "Adoption of electricity in rural rwanda 10 years after connection," *Nature Communications*, vol. 16, no. 1, p. 10 942, Dec. 7, 2025, Publisher: Nature Publishing Group, ISSN: 2041-1723.  DOI: [10.1038/s41467-025-66986-0](https://doi.org/10.1038/s41467-025-66986-0).
- 3 **J. Mugenyi**, G. G. Sutil, and V. Modi, "Electricity consumption: The role of grid reliability in appliance ownership and usage in rwanda," *Energy Economics*, vol. 151, p. 108 907, 2025, ISSN: 0140-9883.  DOI: <https://doi.org/10.1016/j.eneco.2025.108907>.
- 4 **J. Mugenyi**, B. Muhwezi, S. Fobi, C. Massa, J. Taneja, N. J. Williams, and V. Modi, "Post-connection electricity demand and pricing in newly electrified households: Insights from a large-scale dataset in rwanda," *Energy Policy*, vol. 198, p. 114 449, Mar. 1, 2025, ISSN: 0301-4215.  DOI: [10.1016/j.enpol.2024.114449](https://doi.org/10.1016/j.enpol.2024.114449).

Conference Proceedings

- 1 **J. Mugenyi**, H. Natuhweera, and V. Modi, "Patterns, purchases, and policy: Understanding domestic electricity use in rural and peri-urban uganda," presented at the 2025 Africa Electricity Symposium.
- 2 **J. Mugenyi**, I. V. Kinhonhi, and V. Modi, "Electricity outages in uganda: Causes, trends and regional disparities," in *2024 IEEE PES/IAS PowerAfrica*, Oct. 2024, pp. 1–5.  DOI: [10.1109/PowerAfrica61624.2024.10759464](https://doi.org/10.1109/PowerAfrica61624.2024.10759464).
- 3 **J. Mugenyi**, I. V. Kinhonhi, and V. Modi, "Natural language processing reveals core issues in uganda's power grid: A study of outages from 2015-2022," in *2024 IEEE Global Humanitarian Technology Conference (GHTC)*, ISSN: 2473-5728, Oct. 2024, pp. 137–142.  DOI: [10.1109/GHTC62424.2024.10771564](https://doi.org/10.1109/GHTC62424.2024.10771564).
- 4 S. Fobi, **J. Mugenyi**, N. J. Williams, V. Modi, and J. Taneja, "Predicting levels of household electricity consumption in low-access settings," in *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Waikoloa, HI, USA: IEEE, Jan. 2022, pp. 2213–2222, ISBN: 978-1-6654-0915-5.  DOI: [10.1109/WACV51458.2022.00227](https://doi.org/10.1109/WACV51458.2022.00227).
- 5 **J. Mugenyi**, E. Mugume, N. J. Williams, J. Kimani, K. Campbell, E. Uche, and J. Dougherty, "Smart metering technologies for mini grids in africa: An overview," in *2021 IEEE PES/IAS PowerAfrica*, Aug. 2021, pp. 1–5.  DOI: [10.1109/PowerAfrica52236.2021.9543294](https://doi.org/10.1109/PowerAfrica52236.2021.9543294).

Skills

Programming	 Python, R, Java, SQL, L ^A T _E X.
Databases	 MySQL, PostgreSQL, SQLite.
Remote Sensing	 QGIS, ArcGIS, ENVI.
Cloud Computing	 AWS (Lambda, EC2, S3, API Gateway)
Power Systems	 DIGSILENT PowerFactory, HOMER
Methods	 Causal inference, regression analysis, panel data methods, machine learning
Professional	 Technical writing, teaching, stakeholder engagement, research communication.